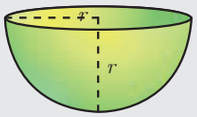
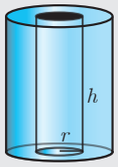
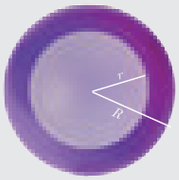
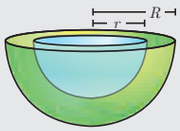
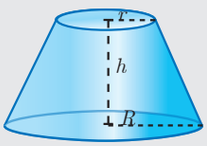
Hemisphere		$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$
Hollow cylinder		$2\pi\left(R+r\right)h$	$2\pi(R+r)$ $(R-r+h)$	$\pi\left(R^2-r^2\right)h$
Hollow sphere		$4\pi R^2$ = outer surface area	$4\pi\left(R^2+r^2\right)$	$\frac{4}{3}\pi\left(R^3-r^3\right)$
Hollow hemisphere		$2\pi\left(R^2+r^2\right)$	$\pi\left(3R^2+r^2\right)$	$\frac{2}{3}\pi\left(R^3-r^3\right)$
Frustum of right circular cone		$\pi\left(R+r\right)l$ where $l=\sqrt{h^2+\left(R-r\right)^2}$	$\pi(R+r)l+\pi R^2$ $+\pi r^2$	$\frac{1}{3}\pi h\left[R^2+r^2+Rr\right]$

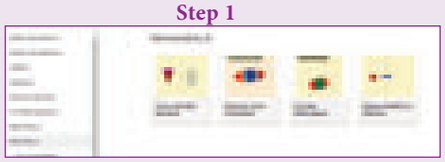
ICT CORNER

ICT 7.1

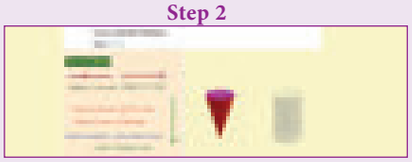
Step 1: Open the Browser type the URL Link given below (or) Scan the QR Code. GeoGebra work book named "Mensuration _X" will open. Select the work sheet "Cone-Cylinder relation"

Step 2: In the given worksheet you can change the radius and height of the cone-Cylinder by moving the sliders on the left-hand side. Move the vertical slider, to view cone filled in the cylinder and this proves 3 times cone equal to one cylinder of same radius and same height.

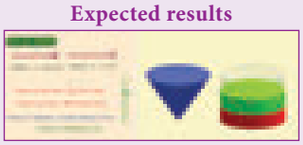
Step 1



Step 2



Expected results



ICT 7.2

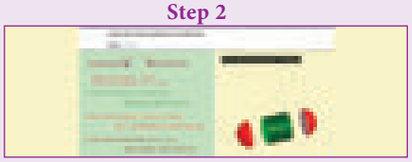
Step 1: Open the Browser type the URL Link given below (or) Scan the QR Code. GeoGebra work book named "Mensuration _X" will open. Select the work sheet "Cylinder Hemispheres"

Step 2: In the given worksheet you can change the radius of the Hemisphere-Cylinder by moving the sliders on the left-hand side. Move the slider Attach/Detach to see how combined solid is formed. You can rotate 3-D picture to see the faces. Working is given on the left-hand side. Work out and verify your answer.

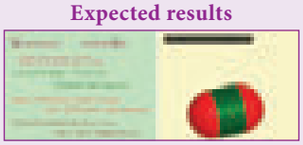
Step 1



Step 2



Expected results



You can repeat the same steps for other activities

https://www.geogebra.org/m/jfr2zzgy#chapter/356197

or Scan the QR Code.

7 Mensuration.indd 300 16-12-2020 03:18:20 PM